Name: _____ **Period:** _____ **Date:** _____

Topic 3.8 Guided Notes: Operant Conditioning

Unit 3: Development and Learning | AP Psychology

Directions: Read through the topic overview page below and fill in each blank as you go. Use the tables and activities to check your understanding, then head to the quiz link at the end to test yourself.

Source page: appsychreviewhub.com/overviews/topic3.8

Word Bank

Fixed Ratio	*Negative Reinforcement*	*Fixed Interval*
Primary Reinforcers	*Continuous Reinforcement*	*Variable Interval*
Negative Punishment	*Variable Ratio*	*Positive Punishment*
Positive Reinforcement	*Instinctive Drift*	*Superstitious Behavior*
Operant Conditioning	*Reinforcement Generalization*	*Law of Effect*
Secondary Reinforcers	*Reinforcement Discrimination*	*Shaping*
Learned Helplessness		

1. The Matrix of Consequences

While classical conditioning is all about involuntary, automatic reflexes (like flinching or salivating), our daily lives are mostly driven by voluntary choices. We study for tests to get good grades, and we obey the speed limit to avoid expensive tickets. This brings us to _____—a type of learning in which behavior is strengthened or weakened based entirely on its consequences.

The foundation of operant conditioning lies in Edward Thorndike's _____, which proposes a simple but powerful rule: behaviors followed by satisfying consequences are more likely to be repeated, while behaviors followed by unpleasant consequences are less likely to occur again.

B.F. Skinner expanded on the Law of Effect by designing highly controlled environments (Skinner boxes) to test how different consequences shaped animal behavior. In operant conditioning, consequences are categorized based on two questions: Are we trying to increase or decrease the behavior? and Are we adding or taking away a stimulus?

Reinforcement (Increasing Behavior)

Reinforcement is any consequence that strengthens a behavior, making it more likely to happen again. These reinforcers can be basic biological needs like food and water (_____) or learned rewards like money and praise that derive their value from being associated with primary needs (_____).

- _____: Adding a desirable stimulus to increase a behavior (e.g., getting a gold star for doing your homework).
- _____: Removing or avoiding an unpleasant stimulus to increase a behavior (e.g., putting on your seatbelt to stop the annoying beeping alarm).

Punishment (Decreasing Behavior)

Punishment is any consequence that weakens a behavior, making it less likely to happen again.

- _____: Adding an unpleasant stimulus to decrease a behavior (e.g., receiving a speeding ticket for driving too fast).
- _____: Removing a desirable stimulus to decrease a behavior (e.g., having your phone taken away for breaking curfew).

Apply It: Which Consequence?

Using the word bank above, identify which concept best matches each scenario.

#	Scenario	Concept
1	An employee receives a $50 bonus for closing an extra sale, making him more likely to pursue extra sales again in the future.	
2	A driver caught running a red light on camera automatically receives a $200 fine in the mail, making her far less likely to run red lights again.	
3	A student who talks back to a teacher during class loses out on recess time for the rest of the week, making her far less likely to talk back again.	
4	An employee who finally finishes an overdue report no longer has to deal with his manager's daily reminder emails, making him more likely to turn in reports on time going forward.	

2. Shaping and Stimulus Control

You cannot simply wait for a rat to accidentally perform a complex trick (like playing a tiny piano) to reward it. Instead, trainers use _____—a process of gradually guiding behavior by rewarding successive approximations of the desired goal until the full behavior is learned.

Just like in classical conditioning, organisms learn to read their environment during operant conditioning:

- _____: The tendency to perform a reinforced behavior in new, similar situations (e.g., a child who gets praised for saying “thank you” at home starts saying it at school).
- _____: The ability to respond only when a specific stimulus is present because that behavior has only been rewarded in that specific situation (e.g., a rat learns that pressing a lever only dispenses food when a green light is on, so it ignores the lever when the light is red).

Apply It: Shaping, Generalization, or Discrimination?

Using the word bank above, identify which concept best matches each scenario.

#	Scenario	Concept
1	A trainer teaches a bird to ring a bell by first rewarding it for approaching the bell, then for pecking near it, and finally only for actually ringing it.	
2	An office worker who gets complimented for wearing a suit to important client meetings starts dressing more formally for regular staff meetings too, even though no one specifically praised that.	

#	Scenario	Concept
3	A dog has learned that begging at the table only earns a treat from Grandpa, so it only begs when Grandpa is at the table and ignores everyone else during dinner.	

3. Schedules of Reinforcement

How often a behavior is rewarded heavily dictates how quickly it is learned and how easily it becomes extinct. The most basic approach is _____, where a behavior is rewarded every single time it occurs. This makes learning incredibly fast, but if the rewards stop, the behavior will extinguish almost immediately.

To create behaviors that are highly resistant to extinction, psychologists use partial reinforcement schedules, which are based either on the number of responses (Ratio) or the passage of time (Interval):

- _____: Rewarding a behavior after a set, predictable number of responses (e.g., getting a free coffee after buying 10 cups). This creates a high rate of responding with a brief pause right after the reward.
- _____: Rewarding a behavior after an unpredictable number of responses (e.g., playing a slot machine). This produces the highest, most consistent rates of behavior and is incredibly hard to extinguish.
- _____: Rewarding the first response after a set amount of time has passed (e.g., studying heavily right before a scheduled Friday quiz). On a graph, this produces a distinct "scalloped" pattern, as behavior spikes right before the time is up.
- _____: Rewarding the first response after unpredictable amounts of time have passed (e.g., checking your phone for a text message). This leads to a slow, steady, and moderate rate of responding.

Apply It: Which Schedule?

Using the word bank above, identify which schedule best matches each scenario.

#	Scenario	Schedule
1	Every single time a toddler puts a toy back in the toy bin, her dad immediately praises her, but the moment he stops praising her for a few days, she quickly stops cleaning up at all.	
2	A factory worker receives a bonus paycheck after every 20 units he successfully assembles, so he tends to work at a fast, steady pace with a short break right after each bonus.	
3	A group of students only crams intensely for their vocabulary quiz in the last hour before it starts each Friday, since they know the quiz always happens on the same weekly schedule.	

4. The Limits of Operant Conditioning

While reinforcement is powerful, it is not absolute. Biology and cognition can interfere with operant learning:

- _____: The biological tendency for an animal to revert to innate, natural behaviors that interfere with a learned conditioned behavior (e.g., a pig trained to drop wooden coins into a piggy bank will eventually start pushing the coins in the dirt with its snout).

- _____: A cognitive state where an organism learns they have no control over negative events, leading them to give up and stop trying to escape, even when a clear opportunity to avoid the punishment is later presented.

Finally, organisms sometimes misinterpret the connection between their actions and consequences. This leads to _____, which occurs when a random consequence accidentally reinforces an entirely unrelated behavior (like wearing "lucky socks" because you happened to ace a test while wearing them once).

📺 *Video Notes: "Review Time" (Classical vs. Operant Conditioning)*

This video reviews classical conditioning (Topic 3.7, up to about 5:44) and then covers operant conditioning (Topic 3.8, from about 5:44). Watch the video above, then answer each question below in your own words.

1. How does the video distinguish classical from operant conditioning, what basic premise does it give for operant conditioning, and how does it describe the Skinner box?

2. Why can't you just wait for a rat to press the lever in a Skinner box, and how does the video say shaping solves that?

3. How does the video define positive reinforcement, negative reinforcement, and punishment, and how does negative reinforcement differ from punishment? Include its examples.

4. What are primary reinforcers and conditioned reinforcers, according to the video? Give its examples.

5. What is continuous reinforcement, what happens when it stops, and why does the video say partial (intermittent) reinforcement matters? What coffee-shop examples does it give?

6. What did behaviorists like Pavlov, Watson, and Skinner believe about how to study psychology, and what did their critics argue?

Apply It: Which Limit?

Using the word bank above, identify which concept best matches each scenario.

#	Scenario	Concept
1	A dog trained to place a ball in a basket for a treat starts repeatedly nudging and pawing at the ball on the ground instead, similar to how it would naturally play with objects before the training.	
2	After repeatedly failing to escape a small locked cage no matter what she tried, a dog stops attempting to escape at all, even after the door is later left wide open.	
3	A baseball player happens to hit a home run while wearing a specific pair of socks, so he insists on wearing those same socks before every single game from then on.	

5. Don't Trip Up! (Common Misconceptions)

⚠️ Negative Reinforcement vs. Punishment: This is arguably the most missed concept on the entire AP Exam! Negative does NOT mean "bad"—it means "subtract" or "take away." _____ removes an annoying or painful stimulus to reward and increase a behavior (like taking an aspirin to remove a headache). Punishment is entirely different because its only goal is to stop a behavior.

6. Quick Check: Review Questions

Circle the best answer for each question below.

1. A man quickly buckles his seatbelt every time he gets in his car specifically because doing so makes the annoying, repetitive beeping alarm stop immediately. Removing this unpleasant beeping sound to increase his seatbelt-buckling behavior is an example of:

- **A.** Negative Punishment
- **B.** Negative Reinforcement
- **C.** Positive Reinforcement
- **D.** Positive Punishment

2. After a teenager breaks his 10 PM curfew, his parents take away his phone privileges for an entire week, and as a result, he becomes much more careful about coming home on time going forward. Removing this desirable phone privilege to decrease his curfew-breaking behavior is an example of:

- **A.** Positive Punishment
- **B.** Negative Punishment
- **C.** Negative Reinforcement
- **D.** Positive Reinforcement

3. A trainer working with a dolphin cannot simply wait for it to accidentally perform a complete, complex jump-and-spin trick on its own. Instead, she first rewards the dolphin for approaching the hoop, then for touching it, then for swimming through it, gradually building up to the full trick over many sessions. This process of rewarding successive approximations toward a final goal behavior is called:

- **A.** Shaping
- **B.** Instinctive Drift
- **C.** Stimulus Generalization
- **D.** Continuous Reinforcement

4. A child who is consistently praised by her parents for saying "thank you" at home begins to spontaneously say "thank you" to her teachers and classmates at school as well, even though she was never specifically rewarded for saying it in that new setting. This tendency to perform a reinforced behavior in new, similar situations is called:

- **A.** Reinforcement Generalization
- **B.** Instinctive Drift
- **C.** Reinforcement Discrimination
- **D.** Shaping

5. A gambler keeps pulling the lever on a slot machine because she never knows exactly how many pulls it will take before she wins again, sometimes winning after just 3 pulls and other times after 50 pulls. This unpredictable reinforcement schedule, which produces the highest and most consistent rates of behavior and is incredibly resistant to extinction, is called:

- **A.** Variable Ratio
- **B.** Fixed Ratio
- **C.** Variable Interval
- **D.** Fixed Interval

6. A teenager checks his phone repeatedly throughout the day at a slow, steady, and fairly consistent rate, since he never knows exactly when a friend might text him back and the wait time varies unpredictably each time. This reinforcement schedule, which rewards the first response after unpredictable amounts of time have passed, is called:

- **A.** Variable Ratio
- **B.** Fixed Ratio
- **C.** Variable Interval
- **D.** Fixed Interval

✅ Check Your Progress

Done with your notes? Head over to the Quiz to test what you've learned:
https://appsychreviewhub.com/quiz

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